

Mantle

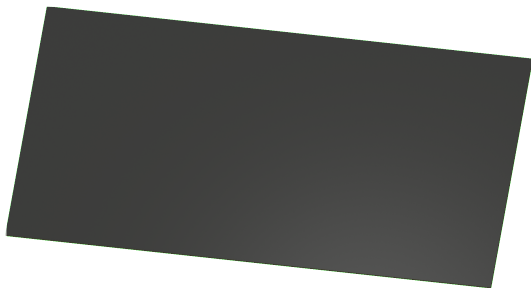
Variant: DRAFT

2026-05-16

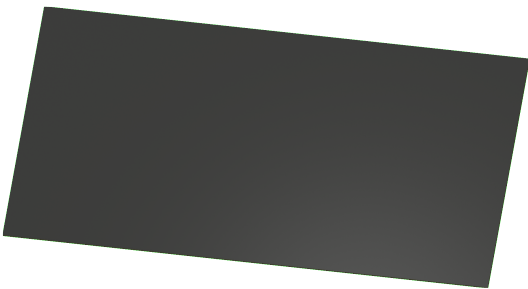
Rev + (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

Date: 16-May-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for informational design notes.

DESIGN NOTE:
Example text for debug notes.

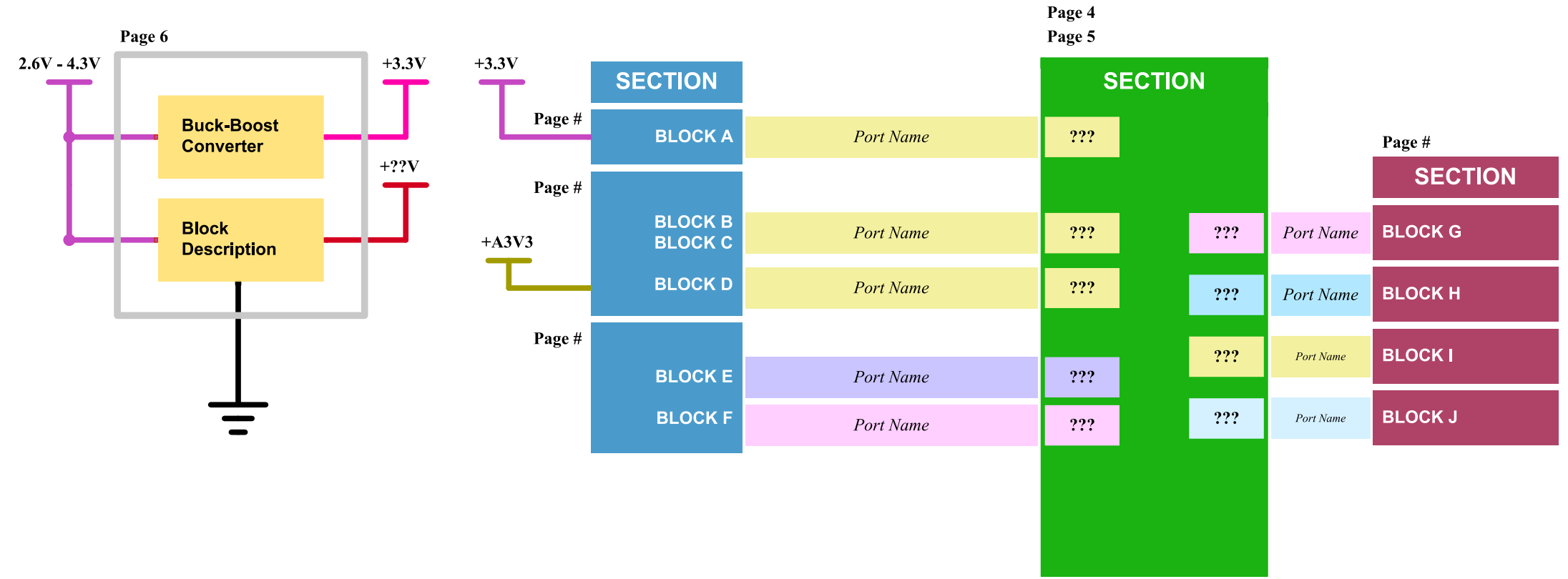
DESIGN NOTE:
Example text for cautionary design notes.

DESIGN NOTE:
Example text for critical design notes.

LAYOUT NOTE:
Example text for critical layout guidelines.

	Comments:	Company: SICEMIC	Variant: DRAFT	Git Hash: 2c18714
	Board Name: Mantle		Project Name: Arctic Seismic Sensor	
	Sheet Title:	File Name: seismic_sensor.kicad_sch	Designer: Arnór Aðalsteinsson	Date: 2025-01-12
	Sheet Path: /		Reviewer:	Revision: + (Unreleased)
			Size: A3	Sheet: 1 of 9

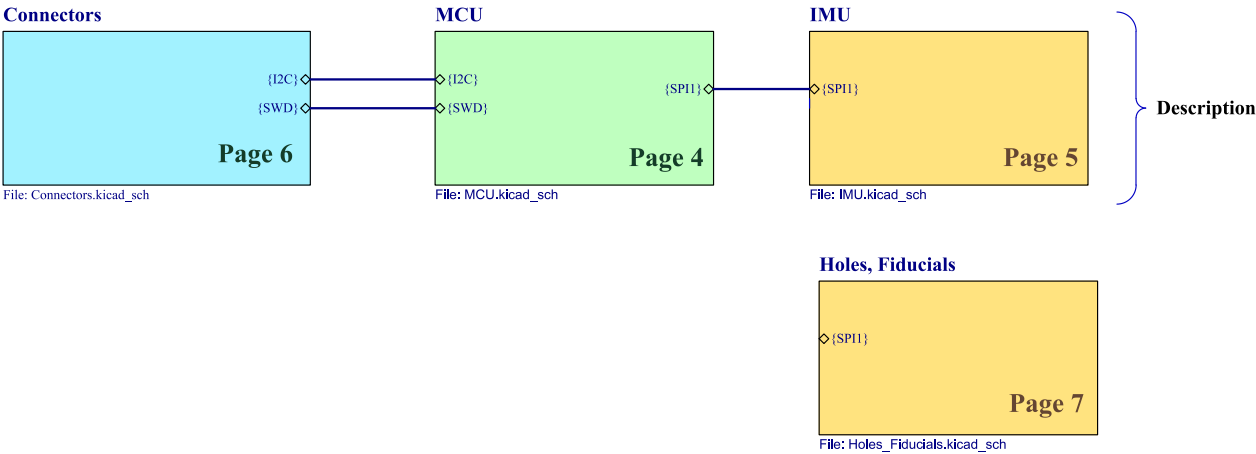
[2] Block Diagram



Target specifications:	
Input voltage:	?? - ?? V
Spec 2	??
Spec 3	??
Spec 4	??

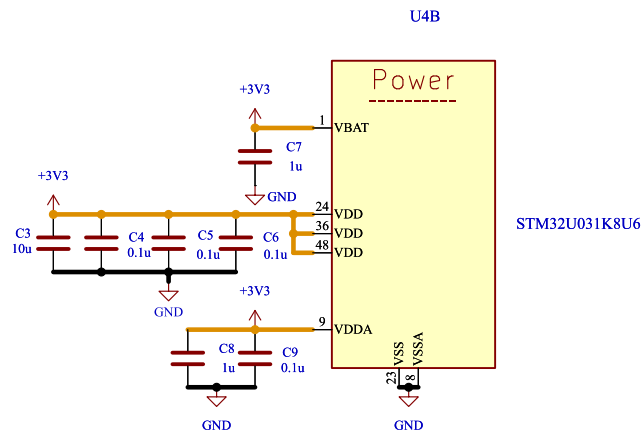
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		SICEMIC		DRAFT		2c18714	
		Board Name:			Project Name:		
		Mantle			Arctic Seismic Sensor		
	Sheet Title:	File Name:		Designer:	Date:	Revision:	
Block Diagram	Block Diagram.kicad_sch		Arnór Aðalsteinsson	2025-01-12	+ (Unreleased)		
Sheet Path:			Reviewer:	Size:	Sheet:		
/Block Diagram/				A3	2 of 9		

[3] Project Architecture



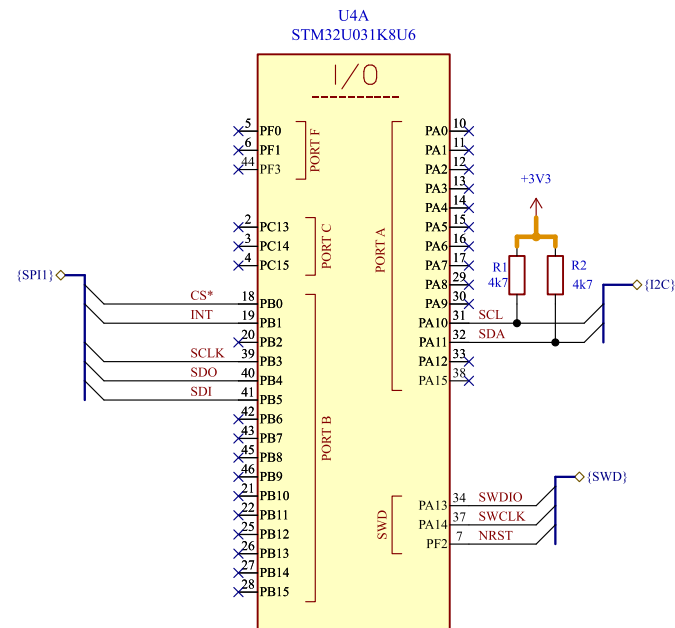
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		Board Name: Mantle		Project Name: Arctic Seismic Sensor	
	Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Arnór Aðalsteinsson	Date: 2025-01-12	Revision: + (Unreleased)
	Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 3 of 9

[4] MCU



LAYOUT NOTE:
biablabla

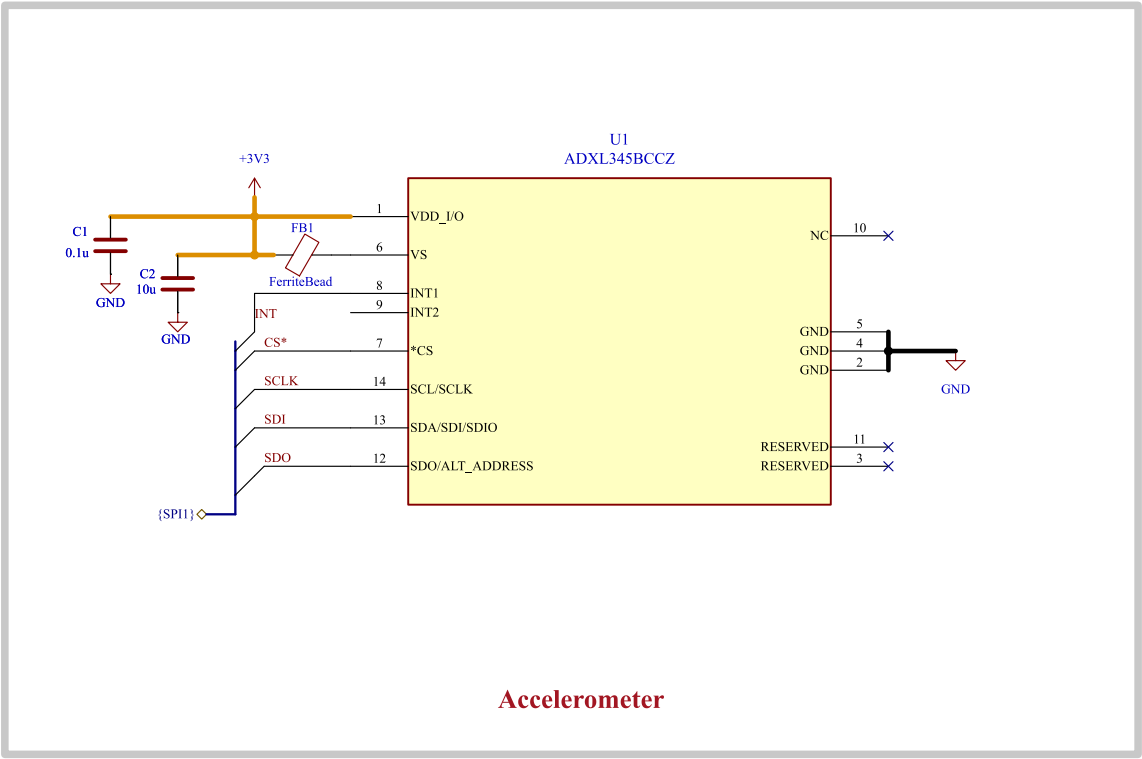
MCU - Power Supply Scheme



MCU - GPIOs

	Comments: AN5346 STM32G474 Datasheet p.81 J. Pieper ADC investigation		Company: SICEMIC		Variant: DRAFT	Git Hash: 2c18714
	Sheet Title: MCU		Board Name: Mantle		Project Name: Arctic Seismic Sensor	
	Sheet Path: /Project Architecture/MCU/		File Name: MCU.kicad_sch	Designer: Arnór Aðalsteinsson	Date: 2025-01-12	Revision: + (Unreleased)
			Reviewer:		Size: A4	Sheet: 4 of 9

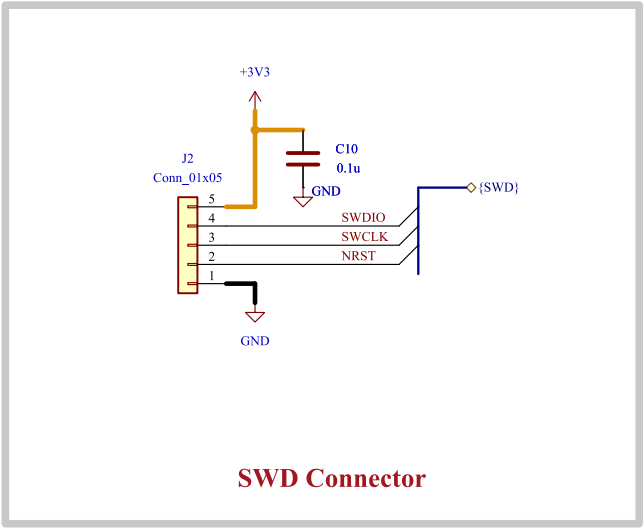
[5] IMU



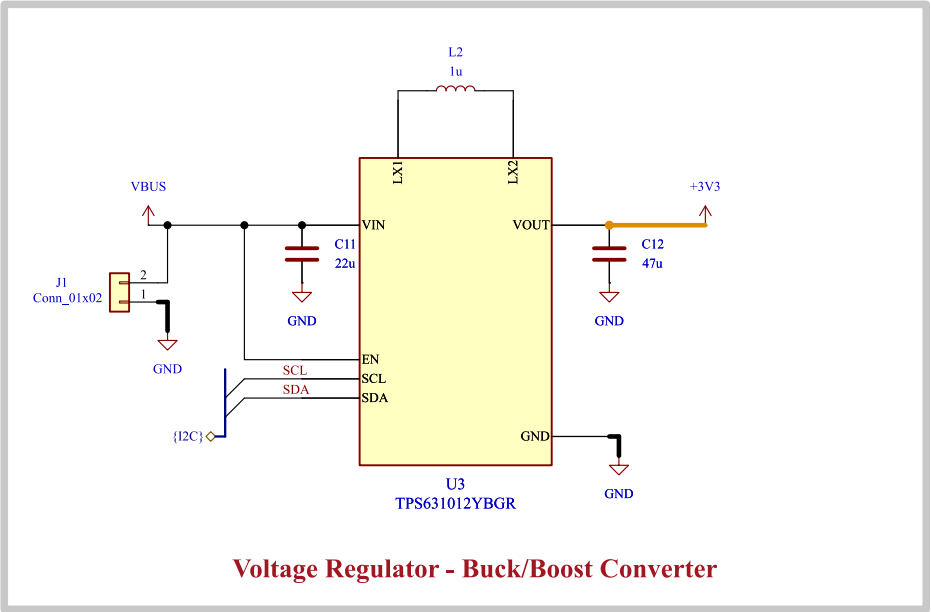
Accelerometer

	Comments:	Company: SICEMIC		Variant: DRAFT	Git Hash: 2c18714
		Board Name: Mantle		Project Name: Arctic Seismic Sensor	
	Sheet Title: IMU	File Name: IMU.kicad_sch	Designer: Arnór Aðalsteinsson	Date: 2025-01-12	Revision: + (Unreleased)
	Sheet Path: /Project Architecture/IMU/		Reviewer:	Size: A4	Sheet: 5 of 9

[6] Connectors



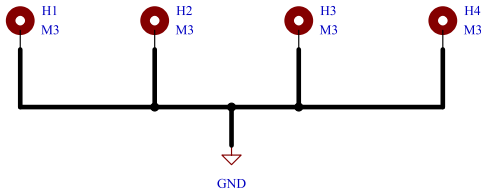
SWD Connector



Voltage Regulator - Buck/Boost Converter

	Comments:	Company: SICEMIC		Variant: DRAFT	Git Hash: 2c18714
		Board Name: Mantle		Project Name: Arctic Seismic Sensor	
	Sheet Title: Connectors	File Name: Connectors.kicad_sch	Designer: Arnór Aðalsteinsson	Date:	Revision: + (Unreleased)
	Sheet Path: /Project Architecture/Connectors/		Reviewer:	Size: A4	Sheet: 6 of 9

[7] Holes, Fiducials



Mounting Holes

Front

- FD1 Fiducial
- FD2 Fiducial
- FD3 Fiducial

Back

- FD4 Fiducial
- FD5 Fiducial
- FD6 Fiducial

Fiducial Markers

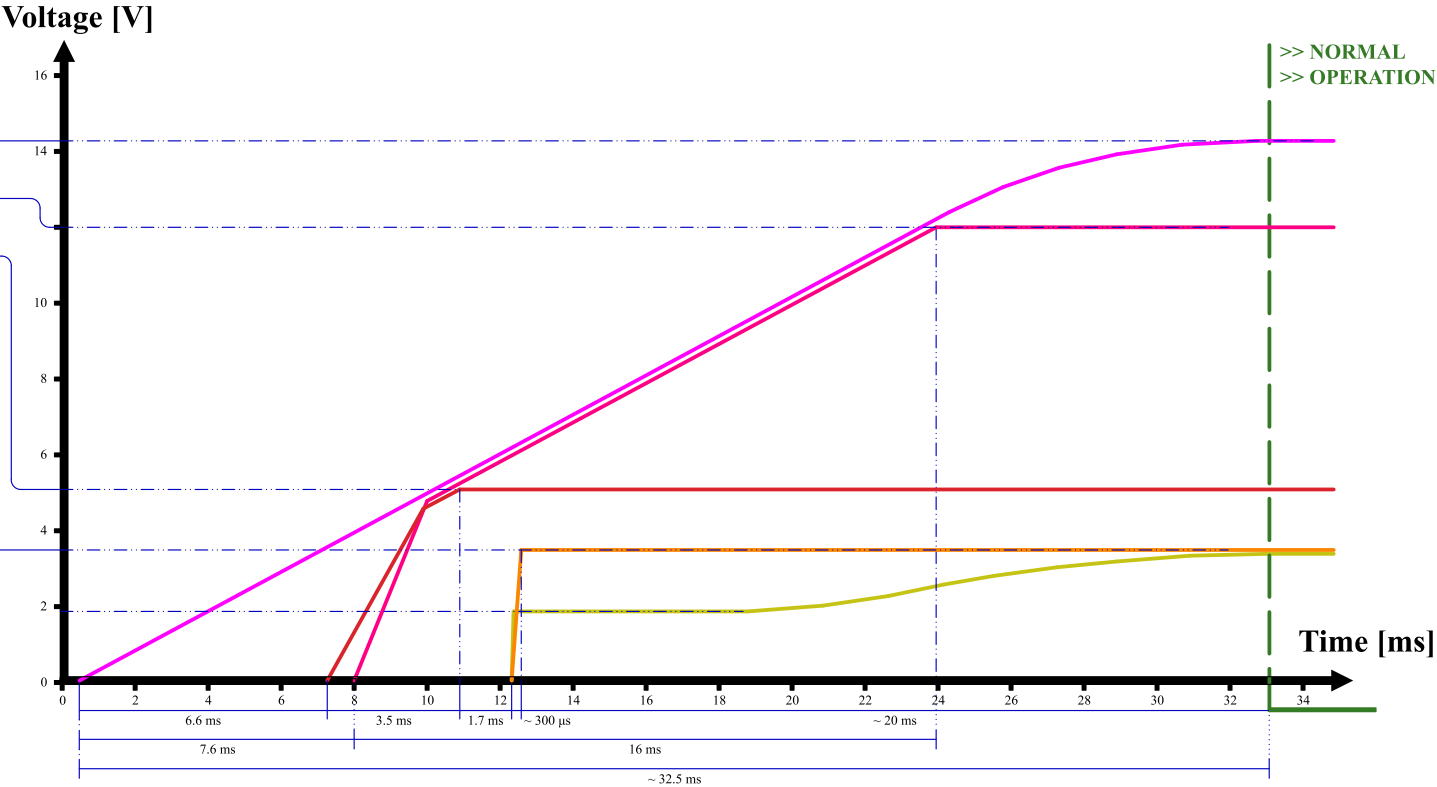
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		Board Name: Mantle		Project Name: Arctic Seismic Sensor	
	Sheet Title: Holes, Fiducials	File Name: Holes_Fiducials.kicad_sch	Designer: Arnór Aðalsteinsson	Date: 2026-02-12	Revision: + (Unreleased)
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[8] Revision History

	Comments:	Company: SICEMIC		Variant: DRAFT	Git Hash: 2c18714
		Board Name: Mantle		Project Name: Arctic Seismic Sensor	
	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Arnór Aðalsteinsson	Date: 2025-01-12	Revision: + (Unreleased)
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 8 of 9

[9] Power - Sequencing

NAME	SOURCE	LEVEL
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%



	Comments:	Company: SICEMIC	Variant: DRAFT	Git Hash: 2c18714
		Board Name: Mantle	Project Name: Arctic Seismic Sensor	
	Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Arnór Aðalsteinsson	Date: 2025-01-12
	Sheet Path: /Power - Sequencing/	Reviewer:	Size: A4	Revision: + (Unreleased)
				Sheet: 9 of 9